

Arka Mukherjee

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- **Research Focus:** LLM reasoning, multimodality, and AI agents

EDUCATION

- **Kalinga Institute of Industrial Technology (KIIT)** Bhubaneswar, India
Bachelor of Technology in Computer Science; Cumulative GPA: 9.74/10.0 *July 2023 - Apr 2027 (Expected)*
 - **Relevant Courses:** Agentic AI, Probability & Statistics, Data Analytics, Machine Learning, AI, Algorithms, Data Mining, Human-Computer Interaction (Coursera)

EXPERIENCE

- **AMD Research and Advanced Development (RAD)** Bengaluru, India
Research Intern *September 2026 - Present*
 - **LLM Inference:** Contributing to LLM inference and decoding techniques and stack to speed up local AI on AMD ROCm GPUs.
- **University of Southern California - Dr. Maja Matarić** Los Angeles, CA
IUSSTF-Viterbi Summer Research Intern *May - August 2026*
 - **Long Video Comprehension:** Built LazySloth, a fast video understanding method that improved retrieval efficiency by up to $8.3\times$ over existing pipelines such as VideoLucy and WorldMM, while retaining task accuracy. Contributed to lab infrastructure on LLM inference. (*Submitted to AAAI 2027*)
- **IIT Bhubaneswar - Dr. Shreya Ghosh** Bhubaneswar, India
Undergraduate Research Fellow (Funded) *December 2024 - Present*
 - **Multimodal STEM reasoning:** Developed mmJEE-Eval, a multimodal reasoning benchmark of 1,460 STEM problems to combat saturation (Qwen 3 VL 235B scored 45% vs 85% on Math Vista). Evaluated 17 VLMs and discovered metacognitive barriers: VLMs detect 53% of errors but correct only 3.5%. (*IJCNLP-AACL 2025*)
 - **Multimodal Cultural Competence:** Created the first systematic **evaluation framework for VLM cultural competence** via story generation on 5 VLMs with a novel evaluation metric. (*Oral @ ICCV 2025 ASI workshop*)
 - **Integrated Capability Evaluation (ICE):** Proposed **ICE**, a new evaluation paradigm unlike benchmarks and leaderboards that models real-world tasks. Engineered CoTDL-mini-SWE-agent, a novel coding harness to improve frontier agentic terminal coding performance. (*Submitted to NeurIPS 2026*)
- **Carnegie Mellon University - Dr. Min Xu** Remote
Research Intern *January - May 2026*
 - **Reinforcement Learning:** Applied GRPO, DPO, and PPO to study RL generalization to codon translation, a low-sampling decoding task (64-token action space). Improved VRAM efficiency of the code to iterate GRPO on 800k+ datapoints.
- **IIT Ropar - Dr. Sudarshan Iyengar** Ropar, India
Summer Research Fellow (IASc-INSANA-SI) *May - July 2025*
 - **VLM Benchmarks:** Developed **EduVLM-Bench** for STEM prerequisite detection and evaluated OSS LLMs. The top model, Gemma3 27B, achieved 38.5% accuracy.

PUBLICATIONS

1. Mukherjee, A., Roy, S., Trivedi, K., & Ghosh, S. (2026). [GeoAgent: Evaluating VLM Geolocalization Through Embodied Navigation](#) (EMNLP 2026 Findings, 14.3% acceptance rate)
2. Mukherjee, A., & Ghosh, S. (2025). [mmJEE-Eval: A Bilingual Multimodal Benchmark for Evaluating Scientific Reasoning in Vision-Language Models](#) (IJCNLP-AACL 2025 Findings, 12% acceptance rate)
3. Mukherjee, A. & Ghosh, S. (2025). [Toward Socially Aware Vision-Language Models: Evaluating Cultural Competence Through Multimodal Story Generation](#) (ICCV 2025 ASI workshop - Oral)

PROJECTS

- **Automated VLM reasoning dataset generator:** Curated 10k samples for SFTing VLMs. Our method achieved the best VMCBench results, beating Walton Cold Start and MM MathInstruct. (*Ranked #6 in NeurIPS DCVLR challenge*)

ACHIEVEMENTS/EXTRACURRICULARS

- **Awards:** **Singapore AI Safety Hub (SASH) Frontier AI Security Training (FAST) fellow** (fully-funded, 2.8% acceptance rate), **KIIT Merit Scholar** (Dean's list) for 6 semesters, **Amgen Scholars** Summer Research Program (declined), **Aalto Science Institute (AScI)** Summer Research Program in Finland (declined)
- **Outreach:** Published articles (10M+ reads) and hardware reviews in collaboration with Nvidia, AMD, and ASUS.
- **Nvidia GPU Sponsorship:** Received RTX 5090, 5080, and 5070 GPUs from Nvidia India for AI research.
- **Travel Awards & Grants:** D&I Subsidy award at IJCNLP-AAACL'25 (\$750), AI Grants India (API keys and credits)

TECHNOLOGIES AND SKILLS

- **Languages:** Python (advanced), Java, C, SQL
- **ML/NLP Frameworks:** PyTorch, HF Transformers, Datasets & TRL, Unsloth, LM Studio, Docker, Agentic AI (LangChain, Playwright, mini-SWE-agent)